

Calculus for Economists

Chain rule, marginal effects, complementarity, and optimisation

Two-hour lecture, extended practice, and complete worked solutions

Running example. A consumer chooses apples $a \geq 0$ and bananas $b \geq 0$, receives utility $U(a, b) = ab$, and faces a budget of £60. Apples cost £3 and bananas cost £6: $3a + 6b \leq 60$.

Learning goals. Differentiate composite and multivariable functions; interpret first and mixed partials; recognise what cross-partial do *not* establish; use non-satiation and indifference curves; solve constrained problems by substitution and by Lagrange multipliers; apply the same ideas to firms.

Suggested timing (120 minutes).

Minutes	Topic
0–15	Ordinary chain rule: functions inside functions
15–30	Partial derivatives and marginal utility
30–47	Second derivatives, mixed partials, complements/substitutes
47–65	Total derivative; budget constraint and substitution
65–83	Indifference curves, MRS, and non-satiation
83–105	Lagrange multipliers, first-order conditions, shadow values
105–120	Practice and applications beyond consumer utility

Teaching choice: Work through the short examples live and assign the longer exercises after each topic or as homework. The answer key begins after the exercises.

1. The chain rule: a function inside another function

If $y = f(g(x))$, then

$$\frac{dy}{dx} = f'(g(x))g'(x).$$

Recipe: differentiate the outside, leave the inside unchanged, and multiply by the derivative of the inside.

$$y = (3x + 2)^4 \implies y' = 4(3x + 2)^3 \cdot 3 = 12(3x + 2)^3.$$

Another example: If revenue is $R(q) = 10q - q^2$ and output is $q(t) = 2t + 1$, then

$$\frac{dR}{dt} = \underbrace{(10 - 2q)}_{\text{marginal revenue}} \underbrace{\frac{dq}{dt}}_{=2} = 2(10 - 2(2t + 1)) = 16 - 8t.$$

Economic intuition. A change in t affects revenue *through* the amount produced.

1.1 How to recognise a composite function

A composite function has one function *inside* another. For example, in $y = (x^2 + 1)^5$, the inside is $g(x) = x^2 + 1$ and the outside is $f(z) = z^5$. Write $z = g(x)$ if you get lost:

$$\frac{dy}{dx} = \underbrace{5z^4}_{dy/dz} \underbrace{2x}_{dz/dx} = 10x(x^2 + 1)^4.$$

Three useful forms to put on the board:

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x), \quad \frac{d}{dx}\sqrt{g(x)} = \frac{g'(x)}{2\sqrt{g(x)}}, \quad \frac{d}{dx}\log(g(x)) = \frac{g'(x)}{g(x)}.$$

Common mistakes: forgetting the inside derivative; differentiating the inside but not the outside; or expanding a complicated power unnecessarily.

1.2 Two more live examples

Nested composite: $y = (1 + (2x + 1)^2)^3$. Working from the outside in,

$$y' = 3(1 + (2x + 1)^2)^2 \cdot 2(2x + 1) \cdot 2.$$

Economics: total cost $C(q) = q^2 + 10$ and production $q(L) = \sqrt{L}$. Thus $C(q(L)) = L + 10$, and the chain rule gives the same result:

$$\frac{dC}{dL} = \frac{dC}{dq} \frac{dq}{dL} = 2\sqrt{L} \frac{1}{2\sqrt{L}} = 1 \quad (L > 0).$$

The chain rule translates a change in an input into a change in an economic outcome.

2. Partial derivatives: changing one variable at a time

For $U(a, b) = ab$, the marginal utilities are

$$MU_a = \frac{\partial U}{\partial a} = b, \quad MU_b = \frac{\partial U}{\partial b} = a.$$

A partial derivative changes a *holding b fixed*, or vice versa. At $(a, b) = (4, 5)$, $MU_a = 5$ and $MU_b = 4$. Because the function is linear in each good separately, adding one apple at fixed $b = 5$ increases utility by exactly 5.

For $U(a, b) = \sqrt{ab} = a^{1/2}b^{1/2}$, at positive bundles,

$$MU_a = \frac{1}{2}a^{-1/2}b^{1/2}, \quad MU_b = \frac{1}{2}a^{1/2}b^{-1/2}.$$

Economic intuition. A marginal effect is not the same as total utility. Compare $U = 20$ with $MU_a = 5$ at $(4, 5)$.

3. Second partials, mixed partials, and economic interactions

Own second derivatives, such as U_{aa} , tell us whether marginal utility of apples rises or falls with more apples. Mixed partials tell us whether *bananas change the marginal utility of apples*:

$$U_{ab} = \frac{\partial}{\partial b} \left(\frac{\partial U}{\partial a} \right), \quad U_{ba} = \frac{\partial}{\partial a} \left(\frac{\partial U}{\partial b} \right).$$

For $U(a, b) = ab$,

$$U_{aa} = U_{bb} = 0, \quad U_{ab} = U_{ba} = 1.$$

Young's theorem (Clairaut's theorem). If second partial derivatives are continuous in a neighbourhood, $U_{ab} = U_{ba}$. This is a mathematical symmetry of derivatives, not an assertion that quantities of apples and bananas must be equal.

Utility	U_{ab}	Interpretation of the cross-partial
ab	1	More bananas raise the marginal utility of apples.
$a + b$	0	Marginal utility of apples does not change with bananas.
$a + b - \frac{1}{2}ab$	$-\frac{1}{2}$	More bananas lower the marginal utility of apples.

Crucial distinction. Positive/negative U_{ab} describes *utility complementarity/substitutability in this representation*, not necessarily whether goods are demand complements/substitutes. The sign can change under an increasing transformation of utility. For example, $U = ab$ and $V = \log a + \log b = \log(ab)$ represent the *same preferences* on $a, b > 0$, but $U_{ab} = 1$ whereas $V_{ab} = 0$. Market-demand substitutes/complements concern how optimal demands respond to *prices*, holding income or utility fixed (depending on the definition).

4 4. The multivariable chain rule and optimisation by substitution

The consumer's budget is $3a + 6b \leq 60$. If they spend all their money, then

$$b(a) = 10 - \frac{a}{2}, \quad \frac{db}{da} = -\frac{1}{2}.$$

When b depends on a , the *total derivative* is

$$\boxed{\frac{dU(a, b(a))}{da} = U_a + U_b \frac{db}{da}.$$

For $U = ab$, this is $b - a/2 = 10 - a$. Equivalently, substitute directly:

$$U(a) = a \left(10 - \frac{a}{2} \right) = 10a - \frac{1}{2}a^2, \quad 0 \leq a \leq 20.$$

First-order condition $10 - a = 0$ gives $a^* = 10$. Then $b^* = 5$, utility $U^* = 50$, and $U''(a) = -1 < 0$. Checking the endpoints ($U(0) = U(20) = 0$) confirms the global maximum.

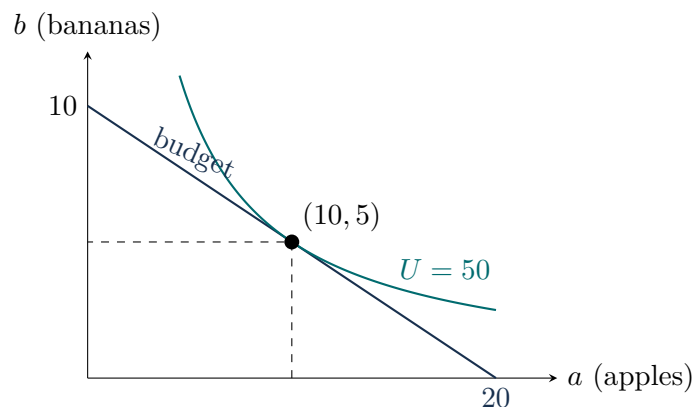
Economic intuition. An extra apple creates direct utility but forces a reduction in bananas. The total derivative combines both changes.

5 5. Indifference curves, trade-offs, and non-satiation

An **indifference curve** is a set of bundles giving equal utility. For $U = ab = \bar{U} > 0$,

$$b = \frac{\bar{U}}{a}, \quad \left. \frac{db}{da} \right|_{U=\bar{U}} = -\frac{U_a}{U_b} = -\frac{b}{a}.$$

The **marginal rate of substitution** (MRS) of apples for bananas is $MRS_{a,b} = U_a/U_b = b/a$: the amount of bananas one would give up for a marginal extra apple while keeping utility unchanged.



Non-satiation. With $U = ab$ and $a, b > 0$, both marginal utilities are positive, so a consumer prefers more of either good. Hence their optimum must use the entire budget: $3a + 6b = 60$. More generally, local non-satiation together with positive prices means a utility-maximiser cannot leave money unspent when more goods are affordable.

Tangency condition. At a smooth *interior* optimum,

$$\boxed{\frac{U_a}{U_b} = \frac{p_a}{p_b} = \frac{3}{6} = \frac{1}{2}} \iff \frac{b}{3} = \frac{a}{6}.$$

At $(10, 5)$, $MRS = 5/10 = 1/2$: the indifference curve and budget line have the same slope. Tangency need not characterise corner optima, and a stationary point by itself need not be a maximum.

6. Lagrange multipliers: solve without eliminating a choice variable

Use a multiplier λ to attach a value to the budget restriction:

$$\mathcal{L}(a, b, \lambda) = ab + \lambda(60 - 3a - 6b).$$

Set the three partial derivatives equal to zero:

$$\frac{\partial \mathcal{L}}{\partial a} = b - 3\lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial b} = a - 6\lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 60 - 3a - 6b = 0.$$

The first two equations give $a = 2b$. Together with the budget, $a^* = 10$, $b^* = 5$, and $\lambda^* = 5/3$.

Why marginal utility per pound equalises:

$$\frac{U_a}{p_a} = \frac{b}{3} = \lambda = \frac{a}{6} = \frac{U_b}{p_b}.$$

If one good produced more marginal utility per pound, the consumer could gain by shifting a little expenditure toward it.

What does λ mean? Let the budget be M instead of 60. The optimum is $a^* = M/6$, $b^* = M/12$, so

$$V(M) = \max_{3a+6b \leq M} ab = \frac{M^2}{72}, \quad V'(M) = \frac{M}{36}.$$

At $M = 60$, $V'(60) = 5/3 = \lambda^*$. Thus an extra pound raises maximum utility by *approximately* $5/3$ units at the original income. The multiplier is a marginal/shadow value, not necessarily the exact gain from an entire extra pound.

When Lagrangians require care. The three equations above treat an interior optimum with a binding equality. If a variable can be zero or a constraint can be slack, use boundary checks or the Kuhn–Tucker conditions.

7. Beyond utility: firms use exactly the same calculus

7.1 Profit maximisation (unconstrained)

A firm's inverse demand is $p(q) = 20 - q$, and cost is $C(q) = 4q + 16$. Profit is

$$\pi(q) = p(q)q - C(q) = 16q - q^2 - 16.$$

$\pi'(q) = 16 - 2q = 0$ implies $q^* = 8$. Since $\pi''(q) = -2$, this is a maximum. *Translation:* marginal revenue equals marginal cost.

7.2 Cost minimisation (constrained)

A firm uses labour L and capital K to produce $q = \sqrt{LK}$, with input prices $w = 2$ and $r = 8$. To produce $q = 10$, minimise $2L + 8K$ subject to $LK = 100$.

$$\mathcal{L} = 2L + 8K + \mu(100 - LK).$$

Conditions $2 - \mu K = 0$, $8 - \mu L = 0$ imply $L = 4K$. With $LK = 100$, the optimum is $L = 20$, $K = 5$, and minimum cost is 80. Substitution ($K = 100/L$) delivers the same answer.

8. Exercises (attempt these before looking at the solutions)

1. **Composite or not?** For each expression, identify the inside and outside functions: (a) $(4x - 1)^6$; (b) $\sqrt{x^2 + 9}$; (c) $(x + 2)(x - 2)$; (d) $\log(3x + 1)$.
2. **Chain rule warm-up.** Differentiate (a) $(5x + 1)^3$; (b) $(7 - 2x)^4$; (c) $(x^2 + 3)^5$.
3. **Square roots and reciprocals.** Differentiate (a) $\sqrt{2x + 5}$; (b) $1/(x^2 + 1)$; (c) $(4 - x^2)^{-2}$ (where defined).
4. **Nested composites.** Differentiate (a) $((x + 1)^2 + 3)^4$; (b) $\sqrt{1 + (3x - 2)^2}$.
5. **Logs and exponentials.** Differentiate (a) $\log(2x^2 + 1)$; (b) $e^{3x^2 - x}$; (c) $\log(\sqrt{x + 1})$ (for $x > -1$).
6. **Spot the missing factor.** A student claims $\frac{d}{dx}(x^2 + 5)^4 = 4(x^2 + 5)^3$. Explain the error and supply the right answer.
7. **Differentiate two ways.** For $y = (x + 1)^2$, find y' using (i) the chain rule and (ii) expansion. Confirm the answers agree.
8. **Revenue via advertising.** Revenue is $R(q) = 20q - q^2$, and advertising t generates sales $q(t) = 2t + 1$. Find dR/dt and evaluate it at $t = 2$. Interpret the sign.
9. **Cost via output.** A firm has $C(q) = 5 + 2q^3$ and output $q(L) = \sqrt{L}$. Calculate dC/dL at $L = 4$ using the chain rule.
10. **Utility from consumption.** A person has $u(c) = \sqrt{c}$ and consumption $c(y) = 0.8y + 4$. Find du/dy at $y = 15$. Give an economic interpretation.
11. **A two-step causal chain.** Price depends on tax t as $p(t) = 10 + t$; demand is $q(p) = 100 - 3p$; revenue is $R(q) = 2q$. Find dR/dt by multiplying the three derivatives.
12. **Challenge: three layers.** Differentiate $y = (1 + (2x^2 + 1)^3)^4$. Indicate all three chain-rule factors.
13. **Ordinary chain rule.** Differentiate (a) $(2x^2 + 1)^3$; (b) $\sqrt{5x + 4}$.
14. **Economic chain rule.** Revenue is $R(q) = 15q - q^2$, while output follows $q(t) = 3t$. Find dR/dt and evaluate at $t = 1$.
15. **Partial derivatives.** For $U(a, b) = a^2b + 3ab^2$, calculate U_a and U_b , and evaluate at $(1, 2)$.
16. **Second derivatives and Young's theorem.** For the same utility, find $U_{aa}, U_{bb}, U_{ab}, U_{ba}$ and verify the mixed partials agree.
17. **Marginal interaction.** For $U(a, b) = a + b - \frac{1}{4}ab$, find U_{ab} . Does this alone establish that the goods are gross substitutes in demand? Explain.
18. **Representation trap.** Calculate U_{ab} for $U = ab$ and V_{ab} for $V = \log a + \log b$ ($a, b > 0$). Do they generate different indifference curves?
19. **Budget and total derivative.** Given $U = ab$ and $4a + 2b = 40$, express b as a function of a , find db/da , and calculate dU/da using partial derivatives.
20. **Substitution.** Solve the previous problem for the optimal a, b . Confirm a maximum, including the endpoints.
21. **Indifference curves.** For $U(a, b) = ab$, derive the slope of an indifference curve. At $(a, b) = (8, 4)$, what is the MRS? What does it mean?

- 22. Non-satiation.** A consumer has $U(a, b) = a + b$, positive prices, and unspent income. Can this be optimal? Would the answer change for $U(a, b) = \min\{a, 5\} + \min\{b, 5\}$ if they already have $(5, 5)$?
- 23. Lagrangian.** Solve $\max ab$ subject to $3a + 6b \leq 120$ with $a, b \geq 0$. Find the multiplier and explain it.
- 24. Different preferences.** Solve $\max a^2b$ subject to $3a + 6b \leq 60$. How is spending divided between the goods?
- 25. Price change.** Solve $\max ab$ subject to $6a + 6b \leq 60$. Compare with the original optimal bundle.
- 26. Firm profit.** A monopolist faces $p(q) = 30 - 2q$ and $C(q) = 6q + 20$. Find the profit-maximising output, price, and profit.
- 27. Firm cost.** Minimise $3L + 12K$ subject to $LK \geq 144$, with $L, K > 0$. Give the optimal input bundle and total cost.
- 28. Challenge: a corner solution.** Maximise $U(a, b) = 3a + b$ subject to $3a + 6b \leq 60$, $a, b \geq 0$. Why would a tangency calculation be misleading?

9. Complete worked solutions

- 1. Solution.** (a) Inside $4x - 1$, outside z^6 ; (b) inside $x^2 + 9$, outside $\sqrt{z}$; (c) a product rather than a nested power, so the natural tool is the product rule (or expansion); (d) inside $3x + 1$, outside $\log z$.
- 2. Solution.** (a) $3(5x + 1)^2 \cdot 5 = 15(5x + 1)^2$; (b) $4(7 - 2x)^3(-2) = -8(7 - 2x)^3$; (c) $5(x^2 + 3)^4(2x) = 10x(x^2 + 3)^4$.
- 3. Solution.** (a) $\frac{1}{2}(2x + 5)^{-1/2}(2) = 1/\sqrt{2x + 5}$; (b) $-(x^2 + 1)^{-2}(2x) = -2x/(x^2 + 1)^2$; (c) $-2(4 - x^2)^{-3}(-2x) = 4x/(4 - x^2)^3$.
- 4. Solution.** (a) $4((x + 1)^2 + 3)^3 \cdot 2(x + 1) = 8(x + 1)((x + 1)^2 + 3)^3$; (b) $\frac{1}{2\sqrt{1+(3x-2)^2}} \cdot 2(3x - 2) \cdot 3 = \frac{3(3x-2)}{\sqrt{1+(3x-2)^2}}$.
- 5. Solution.** (a) $4x/(2x^2 + 1)$; (b) $(6x - 1)e^{3x^2-x}$; (c) applying the chain rule twice gives $[1/\sqrt{x+1}][1/(2\sqrt{x+1})] = 1/[2(x+1)]$.
- 6. Solution.** The student omitted $\frac{d}{dx}(x^2 + 5) = 2x$. Correct: $8x(x^2 + 5)^3$.
- 7. Solution.** Chain rule: $2(x + 1)$. Expansion: $y = x^2 + 2x + 1$, so $y' = 2x + 2 = 2(x + 1)$.
- 8. Solution.** $R_q = 20 - 2q$ and $q_t = 2$, so $R_t = 2[20 - 2(2t + 1)] = 36 - 8t$. At $t = 2$, $R_t = 20 > 0$: a small advertising increase raises revenue at this point (not necessarily profit).
- 9. Solution.** $C_q = 6q^2$, $q_L = 1/(2\sqrt{L})$. At $L = 4$, $q = 2$, so $dC/dL = 6(4)(1/4) = 6$ cost units per extra unit of labour.
- 10. Solution.** $u_c = 1/(2\sqrt{c})$ and $c_y = 0.8$. At $y = 15$, $c = 16$, thus $du/dy = (1/8)(0.8) = 0.1$ utility units per unit of income.
- 11. Solution.** $dR/dq = 2$, $dq/dp = -3$, $dp/dt = 1$. Therefore $dR/dt = 2(-3)(1) = -6$. This is the effect on the specified revenue function; it assumes revenue is $2q$ rather than price times quantity.
- 12. Solution.** Outside to inside: $4(1 + (2x^2 + 1)^3)^3 \cdot 3(2x^2 + 1)^2 \cdot 4x = 48x(1 + (2x^2 + 1)^3)^3(2x^2 + 1)^2$.
- 13. Solution.** (a) $3(2x^2 + 1)^2(4x) = 12x(2x^2 + 1)^2$. (b) $\frac{1}{2}(5x + 4)^{-1/2} \cdot 5 = \frac{5}{2\sqrt{5x+4}}$.
- 14. Solution.** $R'(q) = 15 - 2q$, $q'(t) = 3$. Hence $dR/dt = (15 - 2(3t))3 = 45 - 18t$, which is 27 at $t = 1$.
- 15. Solution.** $U_a = 2ab + 3b^2$, $U_b = a^2 + 6ab$. At $(1, 2)$, $U_a = 4 + 12 = 16$ and $U_b = 1 + 12 = 13$.
- 16. Solution.** $U_{aa} = 2b$, $U_{bb} = 6a$, $U_{ab} = 2a + 6b$, and $U_{ba} = 2a + 6b$. Equality holds; polynomials have continuous second derivatives.
- 17. Solution.** $U_a = 1 - b/4$, so $U_{ab} = -1/4$. No: a negative cross-partial measures how bananas affect the marginal utility of apples in that utility representation. Gross substitutability is a price-demand statement; it cannot be inferred from this sign alone. Also note that this function need not be increasing for all positive bundles.
- 18. Solution.** $U_{ab} = 1$, while $V_a = 1/a$ and $V_{ab} = 0$. They represent the same preferences because $V = \log U$ for $a, b > 0$: each indifference curve $ab = c$ is preserved. This proves that a cross-partial sign is not ordinally invariant.
- 19. Solution.** $b = 20 - 2a$ and $db/da = -2$. The chain rule gives $dU/da = U_a + U_b(db/da) = b - 2a = 20 - 4a$.

- 20. Solution.** $20 - 4a = 0$ gives $a = 5$ and $b = 10$. $U(a) = 20a - 2a^2$, so $U'' = -4 < 0$. At $a = 0$ and $a = 10$ utility is zero, while $U(5, 10) = 50$.
- 21. Solution.** On $ab = \bar{U}$, $b = \bar{U}/a$; differentiation yields $db/da = -\bar{U}/a^2 = -b/a$. The MRS is $b/a = 4/8 = 1/2$: at the margin the consumer gives up half a banana for another apple while maintaining utility.
- 22. Solution.** For $a + b$, any extra affordable good increases utility, so unspent income cannot be optimal. With satiation at $(5, 5)$, additional consumption yields no more utility, so an optimal bundle can leave income unspent. Non-satiation is the key assumption, not merely positive prices.
- 23. Solution.** $\mathcal{L} = ab + \lambda(120 - 3a - 6b)$. FOCs $b = 3\lambda$, $a = 6\lambda$ give $a = 2b$. Thus $b = 10$, $a = 20$, and $\lambda = b/3 = 10/3$. An extra pound of budget raises maximised utility by approximately $10/3$ at income 120.
- 24. Solution.** $\mathcal{L} = a^2b + \lambda(60 - 3a - 6b)$. FOCs: $2ab = 3\lambda$, $a^2 = 6\lambda$. For a positive solution, divide to obtain $2b/a = 1/2$, hence $a = 4b$. Thus $3(4b) + 6b = 60$, $b = 10/3$, $a = 40/3$. Spending: apples £40, bananas £20. Boundary utility is zero and interior utility positive, so this is the maximum.
- 25. Solution.** Symmetry of prices and $U = ab$ gives $a = b$. Then $6a + 6b = 60$ implies $(a, b) = (5, 5)$, rather than original $(10, 5)$. The higher apple price reduces the optimal apple quantity. Bananas happen to stay at 5 for this Cobb–Douglas example.
- 26. Solution.** Revenue $R(q) = 30q - 2q^2$; profit $\pi(q) = 24q - 2q^2 - 20$. FOC $24 - 4q = 0$ gives $q = 6$. Price $p(6) = 18$. Profit $18 \cdot 6 - (6 \cdot 6 + 20) = 52$. The second derivative -4 confirms a maximum.
- 27. Solution.** The production requirement binds: $LK = 144$. $\mathcal{L} = 3L + 12K + \mu(144 - LK)$. FOCs: $3 = \mu K$ and $12 = \mu L$, implying $L = 4K$. Then $4K^2 = 144$, so $K = 6$, $L = 24$. Cost $3(24) + 12(6) = 144$. Both inputs are positive, and substitution gives a strictly convex cost function $3L + 1728/L$ for $L > 0$, ensuring the minimum.
- 28. Solution.** Apples give $3/3 = 1$ utility per pound; bananas give $1/6$ utility per pound. Spend all £60 on apples: $(a, b) = (20, 0)$ and $U = 60$. The optimum is a corner. Since $U_a/U_b = 3$ but $p_a/p_b = 1/2$, the tangency condition does not hold and is not required at a corner.

10 10. Three final takeaways

- Chain rule:** track how an outside outcome changes when an inside variable changes.
- Marginals and cross-marginals:** first derivatives describe marginal effects; second derivatives describe how they change. Cross-partial derivatives are *not* by themselves a demand classification.
- Optimisation:** use the budget and economic assumptions to identify the feasible choices; then use substitution, tangency, or a Lagrangian, checking corners and the nature of the optimum.